NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF APPLIED SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

INTRODUCTION TO COMPUTER SCIENCE AND PROGRAMMING

SCS 1101

Examination Paper

October First Semester Exam 2024

This examination paper consists of 4 pages

Time Allowed:	3 hours
Total Marks:	100
Examiner's Name:	Mr J. Mutengeni
External Examiner:	Dr C. Gombiro

INSTRUCTIONS

1. Answer any four (4) questions
2. Each question carries 25 marks
3. Use of calculators is **not** permissible

MARK ALLOCATION

QUESTION	MARKS
1.	25
2.	25
3.	25
4.	25
5.	25
TOTAL	100

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QUESTION ONE

a) Explain the following terms as they are used in computing.

i. DBMS [2]
ii. IP address [2]
iii. SaaS [2]
iv. Byte [2]
v. Variable(As used in computer Programming) [2]

b) Demonstrate how the following conditional statements are used in C- Programming with the aid of flow charts and code snippets of their implementation

i. if statement [5]

ii. if - else [5]

iii. if else if else if...else [5]

QUESTION TWO

a) Find the Binary equivalent of 156_{Hex} [2]

b) List the three types of errors in computer programming. [3]

c) Evaluate $(BAE)_{hex} + (DAD)_{hex}$ giving your answer in Octal [5]

d) Write a complete C program that adds 10 integer values into an array and calculate their average

[15]

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QUESTION THREE

a) The NUST Computer Science department is having a sixteen-station Local Area Network (LAN) installed in one of their Labs.

i. Describe what is meant by the term Local Area Network. [2]

ii. Give two suitable topologies for the LAN and, for each one, draw a labelled diagram to show its structure. [8]

b) ZESA charges the following rates for the use of electricity: for the first 200 units 80 cents per unit: for the next 100 units 90 cents per unit: beyond 300 units 1 USD per unit. All users are charged a minimum of $10 as meter charge. If the total amount is more than $400, then an additional surcharge of 15% of total amount is charged. Write a program to read the number of units consumed and print out the charges. [15]

QUESTION FOUR

a) List any five functions of the operating System [5]

b) Identify the errors in the following program

```
1. #Include <stdio.h>
2. /* run this program using the console pauser or add your own getch,
   system("pause") or input loop
3. int main[ ] (
4. float x
5. X=5.4;
6. printf(" Welcome to C Programming );
7. printf("%d\n" ,x);
8. print f("%f ",x+=2.5);
9. return 0. }
```

[10]

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c) Compare and contrast Databases and Cloud Computing [10]

QUESTION FIVE

a) Parameters can be passed by value or by reference in a function, Explain the difference you may use code snippets in your explanation [5]

b) Explain any five advantages of using cloud storage [10]

c) Write a C Program that uses a recursive function to generate the Fibonacci series [10]

END OF QUESTION PAPER

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